

Supplementary Material: Leakage-Controlled Concept Graph Construction and Cold-Start Diagnostic Protocol for Knowledge Tracing

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Abstract

This supplement provides per-dataset baseline and ablation tables, paired significance tests, full cold-start strata, DDR downstream scatter plot, sequence-autocorrelation statistics, operator definitions, ground-truth disagreement examples, extended ground-truth metrics with node Jaccard, and supplementary DDR / KT-graph figures omitted from the main manuscript for length. Tables S15–S18 report training parity, Δ AUC intervals (paired- t by default), controlled leak-injection metrics, and downstream AUC on injected graphs. Table S21 reports the GKT epoch-extended ablation on XES3G5M; Table S22 reports multi-seed replication.

Supplementary index

Tables. S1–S5: per-dataset baseline diagnostics (Section 1); S6–S10: per-dataset graph ablation (Section 2); S11: paired significance tests on public benchmarks (Section 3); S12–S13: cold-start diagnostics (Section 8); S14: ground-truth metrics including node

Table 1: Diagnostic baseline results (train-only graphs). Baselines are used for protocol comparison only; no SOTA claim is made. Values are fold means with 95% bootstrap confidence intervals over valid+test sequence positions (`n_folds` as exported per dataset).

Model	AUC		ACC		NLL	
AKT	0.984	[0.982, 0.989]	0.951	[0.945, 0.962]	0.121	[0.093, 0.135]
BKT	0.780	[0.779, 0.781]	0.846	[0.845, 0.847]	0.379	[0.377, 0.380]
DGEKT	0.981	[0.981, 0.981]	0.943	[0.942, 0.943]	0.137	[0.136, 0.138]
DKT	0.984	[0.982, 0.989]	0.951	[0.945, 0.962]	0.120	[0.093, 0.134]
DyGKT	0.978	[0.978, 0.978]	0.934	[0.933, 0.934]	0.151	[0.151, 0.152]
GIKT	0.982	[0.982, 0.982]	0.946	[0.946, 0.946]	0.133	[0.132, 0.133]
<i>simpleKT</i>	0.984	[0.982, 0.989]	0.951	[0.945, 0.962]	0.121	[0.093, 0.135]
SKT	0.979	[0.976, 0.986]	0.938	[0.931, 0.954]	0.139	[0.107, 0.156]

Jaccard (Section 7); S15: training parity (Section 11); S16: Δ AUC intervals for primary pairs (Section 12); S17–S18: controlled leak injection and downstream AUC (Section 15); S19–S20: exploratory ANOVA on backbone and DGEKT downstream AUC (Section 16); S21: GKT epoch-extended ablation, seed 42 (Section 13); S22: multi-seed GKT 30ep replication (Section 14).

Sections and figures. Section S3: sequence-autocorrelation analysis (Section 4); Section S4: DDR augmentation operators (Section 5); Section S5: ground-truth disagreement examples (Section 6); Figure S1: multi-benchmark train-only E_{pre} samples (Figure 3); Figure S2: multi-benchmark DDR curves (Figure 4); Figure S3: DDR downstream scatter plot (Figure 2). Cold-start panels in the main manuscript (Figure 7) are generated from fold 0 exports by `scripts/plot\_kt\_graph\_figures.py`.

1 Per-dataset baseline diagnostics

Tables S1–S5 (Tables 1–5) reproduce the five diagnostic baseline tables (AUC, ACC, NLL with 95% bootstrap CIs) that the main manuscript summarises in its cross-dataset AUC summary table. pyKT GKT is omitted on Junyi Academy because the stock wrapper is impractical on the dense multi-million-interaction graphs under our compute budget.

2 Per-dataset graph construction ablation

Tables S6–S10 (Table 6–10) report fold 0 train-only versus full-log metrics for each graph-augmented pyKT model. The main manuscript reports only the maximum $|\Delta$ AUC— and $|\Delta$ ACC per dataset.

3 Paired significance tests (public benchmarks)

Table S11 (Table 11) reports paired t -test and Wilcoxon p -values for graph-augmented models versus *simpleKT* on the three public benchmarks. The main manuscript

Table 2: Diagnostic baselines: ASSISTments 2012 (three-fold learner CV (`n_folds=3`); train-only graphs).

Model	AUC		ACC		NLL	
AKT	0.968	[0.968, 0.969]	0.916	[0.914, 0.917]	0.175	[0.173, 0.176]
BKT	0.669	[0.667, 0.671]	0.713	[0.711, 0.715]	0.574	[0.573, 0.576]
DGEKT	0.960	[0.959, 0.961]	0.907	[0.906, 0.908]	0.186	[0.185, 0.188]
DKT	0.963	[0.961, 0.964]	0.910	[0.908, 0.911]	0.182	[0.180, 0.185]
DyGKT	0.952	[0.951, 0.952]	0.899	[0.898, 0.900]	0.200	[0.198, 0.203]
GIKT	0.967	[0.967, 0.968]	0.915	[0.914, 0.915]	0.175	[0.174, 0.177]
GKT	0.961	[0.960, 0.962]	0.907	[0.906, 0.908]	0.185	[0.184, 0.187]
<i>simpleKT</i>	0.968	[0.967, 0.968]	0.916	[0.915, 0.916]	0.173	[0.172, 0.175]
SKT	0.953	[0.952, 0.953]	0.898	[0.897, 0.899]	0.198	[0.196, 0.200]

Table 3: Diagnostic baselines: XES3G5M (three-fold learner CV (`n_folds=3`); train-only graphs).

Model	AUC		ACC		NLL	
AKT	0.875	[0.874, 0.875]	0.858	[0.857, 0.858]	0.324	[0.323, 0.326]
BKT	0.636	[0.634, 0.638]	0.793	[0.791, 0.793]	0.490	[0.490, 0.492]
DGEKT	0.841	[0.841, 0.842]	0.844	[0.844, 0.845]	0.358	[0.356, 0.358]
DKT	0.858	[0.857, 0.858]	0.851	[0.851, 0.852]	0.342	[0.341, 0.342]
DyGKT	0.752	[0.752, 0.754]	0.813	[0.813, 0.814]	0.431	[0.429, 0.431]
GIKT	0.878	[0.878, 0.878]	0.860	[0.859, 0.860]	0.321	[0.320, 0.322]
GKT	0.834	[0.833, 0.835]	0.842	[0.841, 0.842]	0.364	[0.363, 0.365]
<i>simpleKT</i>	0.875	[0.874, 0.875]	0.858	[0.857, 0.859]	0.324	[0.323, 0.325]
SKT	0.625	[0.623, 0.626]	0.800	[0.799, 0.801]	0.482	[0.480, 0.483]

baseline-CV table lists only Wilcoxon p -values because three-fold CV limits Wilcoxon resolution ($p_{\min}=0.25$).

4 Sequence autocorrelation analysis

For every learner we order interactions by timestamp and, over consecutive within-learner pairs $(t-1, t)$, measure KC-repeat rate, item-repeat rate, label persistence, lag-one label autocorrelation ρ , and the AUC of the trivial previous-answer predictor (`scripts/compute_autocorrelation.py`).

5 DDR augmentation operator definitions

All operators act on the retained acyclic E_{pre} at perturbation strength $p \in \{0.05, 0.10, 0.20, 0.30\}$:

- **Attribute masking** masks a random subset of edge attributes without changing E_{pre} (DDR=0 by construction).
- **Edge dropping** removes each edge independently with probability p (expected DDR $\approx p$).

Table 4: Diagnostic baselines: Synthetic C2 (three-fold learner CV (`n_folds=3`); train-only graphs).

Model	AUC		ACC		NLL	
AKT	0.873	[0.870, 0.876]	0.808	[0.802, 0.812]	0.407	[0.401, 0.417]
BKT	0.835	[0.833, 0.838]	0.782	[0.780, 0.784]	0.460	[0.458, 0.462]
DGEKT	0.710	[0.634, 0.859]	0.724	[0.686, 0.794]	0.551	[0.432, 0.613]
DKT	0.862	[0.857, 0.868]	0.798	[0.788, 0.805]	0.421	[0.413, 0.434]
DyGKT	0.710	[0.631, 0.867]	0.726	[0.686, 0.801]	0.550	[0.421, 0.617]
GIKT	0.875	[0.871, 0.880]	0.809	[0.802, 0.815]	0.405	[0.398, 0.416]
GKT	0.873	[0.868, 0.878]	0.808	[0.799, 0.814]	0.407	[0.400, 0.420]
<i>simpleKT</i>	0.875	[0.871, 0.880]	0.809	[0.803, 0.814]	0.405	[0.399, 0.416]
SKT	0.708	[0.634, 0.853]	0.722	[0.686, 0.788]	0.554	[0.440, 0.613]

Table 5: Diagnostic baselines: Synthetic C5 (three-fold learner CV (`n_folds=3`); train-only graphs).

Model	AUC		ACC		NLL	
AKT	0.822	[0.819, 0.823]	0.750	[0.746, 0.752]	0.500	[0.496, 0.507]
BKT	0.791	[0.789, 0.794]	0.724	[0.722, 0.726]	0.534	[0.532, 0.536]
DGEKT	0.830	[0.827, 0.832]	0.756	[0.755, 0.757]	0.489	[0.486, 0.492]
DKT	0.826	[0.825, 0.827]	0.754	[0.752, 0.756]	0.493	[0.491, 0.496]
DyGKT	0.826	[0.824, 0.828]	0.751	[0.748, 0.753]	0.492	[0.490, 0.496]
GIKT	0.820	[0.817, 0.824]	0.749	[0.744, 0.754]	0.500	[0.496, 0.504]
GKT	0.810	[0.806, 0.814]	0.738	[0.730, 0.744]	0.526	[0.513, 0.538]
<i>simpleKT</i>	0.829	[0.828, 0.831]	0.755	[0.754, 0.757]	0.491	[0.489, 0.492]
SKT	0.810	[0.808, 0.813]	0.740	[0.737, 0.744]	0.511	[0.507, 0.513]

Table 6: Ablation contrasting *train-only* graph smoothing (protocol) with a *full-log* graph built from all learners before splitting (leaky construction). Graph-augmented rows are pyKT checkpoints consuming exported adjacency from P0. Δ is fold 0 full-log minus train-only. *to*: train-only graph; *fl*: full-log graph.

Model	AUC to	AUC fl	Δ AUC	ACC to	ACC fl	Δ ACC
GIKT	0.982	0.982	0.000	0.946	0.946	-0.000
SKT	0.979	0.976	-0.003	0.938	0.931	-0.008
DyGKT	0.978	0.978	-0.000	0.934	0.934	0.000
DGEKT	0.981	0.981	-0.000	0.943	0.943	-0.000

- **Subgraph sampling** performs a random walk until $\lceil(1-p)|\mathcal{C}|\rceil$ nodes are visited and returns the induced subgraph (boundary edges drop; internal connectivity preserved).
- **Node dropping** removes $\lceil p|\mathcal{C}|\rceil$ nodes and all incident edges.
- **prereq_preserve** draws the same per-edge budget as edge dropping but removes only transitively redundant edges, protecting the transitive-reduction backbone.

Table 7: Train-only vs. full-log graph ablation: ASSISTments 2012.

Model	AUC to	AUC fl	Δ AUC	ACC to	ACC fl	Δ ACC
GKT	0.961	0.961	-0.000	0.907	0.907	-0.000
GIKT	0.967	0.968	0.000	0.915	0.915	0.000
SKT	0.953	0.953	0.000	0.898	0.898	0.000
DyGKT	0.952	0.952	-0.000	0.899	0.899	-0.000
DGEKT	0.960	0.960	0.000	0.907	0.907	0.000

Table 8: Train-only vs. full-log graph ablation: XES3G5M.

Model	AUC to	AUC fl	Δ AUC	ACC to	ACC fl	Δ ACC
GKT	0.834	0.835	0.002	0.842	0.842	0.000
GIKT	0.878	0.878	0.000	0.860	0.860	0.000
SKT	0.625	0.624	-0.000	0.800	0.800	0.000
DyGKT	0.752	0.749	-0.003	0.813	0.812	-0.001
DGEKT	0.841	0.841	-0.000	0.844	0.844	-0.001

Table 9: Train-only vs. full-log graph ablation: Synthetic C2.

Model	AUC to	AUC fl	Δ AUC	ACC to	ACC fl	Δ ACC
GKT	0.873	0.873	-0.001	0.808	0.807	-0.001
GIKT	0.875	0.874	-0.000	0.809	0.809	0.000
SKT	0.708	0.708	0.000	0.722	0.723	0.001
DyGKT	0.710	0.710	0.000	0.726	0.726	-0.000
DGEKT	0.710	0.711	0.001	0.724	0.725	0.001

6 Ground-truth disagreement examples (Junyi)

The cross-validation procedure exports qualitative disagreement cases to `results/tables/disagreement_examples.csv`. Inferred-only edges are predominantly bidirectional co-occurring arithmetic pairs (e.g. `number_line` $\leftrightarrow$ `addition_1`). Expert-only edges are systematic level-progression annotations (e.g. `equivalent_fractions_2` $\rightarrow$ `equivalent_fractions`). The `wrong_direction` category contains many reversed level-progression pairs such as `addition_1` $\rightarrow$ `addition_2`.

7 Ground-truth metrics with node Jaccard

Table S14 (Table 13) extends the main manuscript ground-truth table with node Jaccard across the same top- K cutoffs.

Table 10: Train-only vs. full-log graph ablation: Synthetic C5.

Model	AUC to	AUC fl	Δ AUC	ACC to	ACC fl	Δ ACC
GKT	0.810	0.812	0.002	0.738	0.738	0.000
GIKT	0.820	0.820	-0.000	0.749	0.747	-0.002
SKT	0.810	0.809	-0.001	0.740	0.739	-0.001
DyGKT	0.826	0.826	0.000	0.751	0.751	0.001
DGEKT	0.830	0.830	0.000	0.756	0.756	-0.001

Table 11: Paired comparisons against *simpleKT* on public benchmarks (three-fold learner CV). Δ is mean AUC difference (model minus *simpleKT*); fold-level magnitudes in the main baseline-CV table are the primary evidence. p_t and p_W are reported for transparency only (underpowered at $n=3$ folds).

Dataset	Model	Δ AUC	p_t	p_W
ASSISTments 2012	GKT	-0.0069	<0.001	0.25
	GIKT	-0.0005	0.001	0.25
	SKT	-0.0154	<0.001	0.25
	DyGKT	-0.0160	<0.001	0.25
	DGEKT	-0.0079	<0.001	0.25
XES3G5M	GKT	-0.0411	<0.001	0.25
	GIKT	+0.0031	<0.001	0.25
	SKT	-0.2501	<0.001	0.25
	DyGKT	-0.1223	<0.001	0.25
	DGEKT	-0.0334	<0.001	0.25
Junyi Academy	GIKT	-0.0021	0.445	1.00
	SKT	-0.0055	0.031	0.25
	DyGKT	-0.0066	0.100	0.25
	DGEKT	-0.0034	0.267	0.25

Table 12: Sequence-autocorrelation diagnostics on the three public benchmarks. *KC rep.* and *Item rep.* are the fractions of consecutive within-learner interactions that repeat the previous KC or item; *Persist. acc.* is $P(y_t=y_{t-1})$; *Lag-1 ρ* is the lag-one label autocorrelation; *Prev-AUC* is the AUC of the trivial predictor $\hat{y}_t=y_{t-1}$.

Dataset	KC rep.	Item rep.	Persist. acc.	Lag-1 ρ	Prev-AUC
Junyi Academy	0.755	0.755	0.786	0.244	0.622
ASSISTments 2012	0.662	0.000	0.664	0.207	0.604
XES3G5M	0.211	0.145	0.786	0.342	0.671

8 Cold-start tables (full)

Table S12 (Table 14) reports fold 0 *simpleKT* metrics by KC-frequency stratum; Table S13 (Table 15) lists three-fold per-model AUC (mean $\pm$ std) and $\Delta_{\max}$ by stratum. The main manuscript summarises both in a single table.

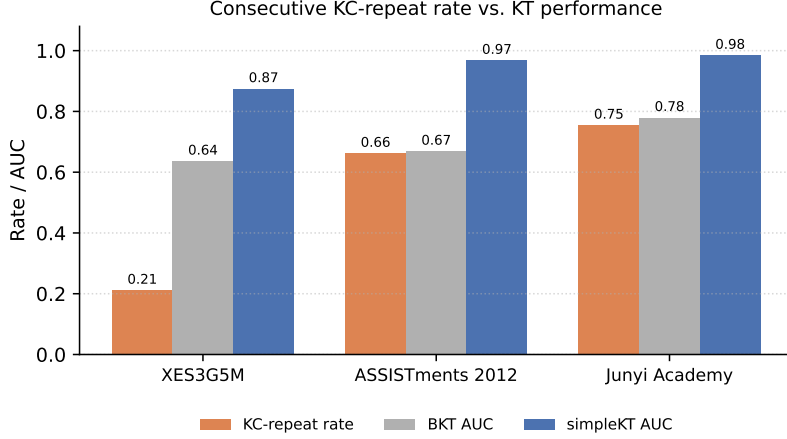


Fig. 1: Consecutive KC-repeat rate against *simpleKT* AUC across the three public benchmarks.

Table 13: Ground-truth cross-validation on Junyi: edge/direction/reachability metrics and node Jaccard across top- K cutoffs. The main manuscript table reports edge precision/recall, direction agreement, and reachability F1 only.

K	Edge F1	Dir. agr.	Rch. F1	Node Jaccard
200	0.071	0.346	0.008	0.290
500	0.109	0.225	0.021	0.453
800	0.150	0.280	0.062	0.572
$ \text{expert} =1131$	0.183	0.261	0.104	0.632
1200	0.185	0.262	0.126	0.641
1800	0.188	0.278	0.260	0.644
2500	0.187	0.285	0.339	0.610
5000 (cap)	0.187	0.285	0.339	0.610

9 DDR downstream scatter plot

10 Additional KT-graph figures

11 Training parity

Table S15 (Table 16) outlines hyperparameter parity and model configuration across pyKT and native baselines. Settings shared by all models are kept out of the table: learning rate 0.001 (Adam), zero hyperparameter-search budget, split seed 42, identical processed splits under both train-only and full-log graph constructions, and checkpoint selection by best validation AUC with early stopping (BKT is fit by L-BFGS on NLL without early stopping). Epoch caps and batch sizes differ across families (30 epochs for sequence checkpoints vs. 10 for graph backbones in the primary release), reflecting

Table 14: Per-stratum cold-start KC diagnostic with *simpleKT* on fold 0. The n column counts test interactions whose KC falls in the stratum train-frequency bin; *out of range* marks test KCs outside the four canonical bins. Stratum mass varies by dataset split; under default bins Synthetic C2/C5 route all test interactions into the hot stratum (> 500 train-fold hits per knowledge component).

Dataset	Stratum	n	AUC	ACC	NLL
Junyi Academy	very_cold	19	1.000	0.684	0.456
	cold	256	0.447	0.621	0.633
	warm	11,257	0.615	0.727	0.572
	hot	7,725,588	0.718	0.831	0.413
	out of range	10	0.500	0.900	0.511
ASSIST'12	very_cold	49	0.608	0.755	0.631
	cold	707	0.769	0.717	0.565
	warm	9,017	0.765	0.687	0.587
	hot	807,178	0.693	0.704	0.574
XES3G5M	very_cold	161	0.810	0.814	0.424
	cold	2,592	0.798	0.794	0.453
	warm	17,655	0.790	0.805	0.431
	hot	1,902,424	0.717	0.796	0.467
	out of range	6	0.625	0.667	0.638
Synthetic C2	hot	60,000	0.624	0.683	0.607
Synthetic C5	hot	60,000	0.630	0.611	0.651

per-epoch cost. Tables S21–S22 report GKT epoch-extended ablations (30 epochs, batch 32; *simpleKT* reference at 10 epochs, batch 64) with fold-aligned graph rebuilds per split seed (17, 42, 1234). “Native” denotes our re-implementations in `src/models`; graph models consume the exported E_{pre} and, where available, E_{sim} edges.

12 ΔAUC intervals for primary pairs

Table S16 (Table 17) reports the mean AUC difference (ΔAUC) between the primary model pairs on XES3G5M under train-only graph construction. The released bundle reports paired- t 95% intervals over three learner-disjoint folds (default; matches fold-level ΔAUC). Optional row-level learner bootstrap: `scripts/bootstrap_auc_ci.py--learner-bootstrap`. The GKT deficit under the released 10-epoch budget (-0.041 ; CI $[-0.044, -0.038]$) is more than an order of magnitude larger than the interval half-width, so the model-specific conclusion does not hinge on fragile three-fold t -tests. Tables S21–S22 show GKT epoch extension recovers only $\approx +0.003$ mean AUC on seed 42, leaving pooled $\Delta \approx -0.038$ (95% CI $[-0.041, -0.035]$ over nine folds). Reproducible may optionally run `python scripts/bootstrap_auc_ci.py--learner-bootstrap` on exported prediction parquets for row-level learner resampling (slower; fold-level paired- t is the default).

Table 15: Cold-start comparison of sequence-only (No-Graph) and graph-aware (Graph) KT models. Entries are three-fold mean $\pm$ std AUC by KC train-frequency stratum; $\Delta_{\max}$ is the best Graph AUC minus the best No-Graph AUC in the same dataset/stratum.

Dataset	Stratum	No-Graph		Graph			
		DKT	SimpleKT	GKT	GIKT	DGEKT	$\Delta_{\max}$
assist2012	very_cold	0.723 $\pm$ 0.085	0.724 $\pm$ 0.100	0.695 $\pm$ 0.100	0.726 $\pm$ 0.102	0.724 $\pm$ 0.100	+0.002 $\pm$ 0.143
		0.769 $\pm$ 0.004	0.772 $\pm$ 0.003	0.750 $\pm$ 0.013	0.772 $\pm$ 0.003	0.772 $\pm$ 0.003	+0.000 $\pm$ 0.013
	cold	0.751 $\pm$ 0.016	0.756 $\pm$ 0.017	0.724 $\pm$ 0.015	0.757 $\pm$ 0.017	0.756 $\pm$ 0.017	+0.001 $\pm$ 0.024
		0.687 $\pm$ 0.002	0.696 $\pm$ 0.002	0.601 $\pm$ 0.001	0.691 $\pm$ 0.002	0.687 $\pm$ 0.002	-0.005 $\pm$ 0.003
	warm	0.813 $\pm$ 0.324	0.813 $\pm$ 0.324	0.813 $\pm$ 0.324	0.813 $\pm$ 0.324	0.813 $\pm$ 0.324	+0.000 $\pm$ 0.458
		0.505 $\pm$ 0.239	0.505 $\pm$ 0.239	0.505 $\pm$ 0.239	0.505 $\pm$ 0.239	0.505 $\pm$ 0.239	+0.000 $\pm$ 0.338
	hot	0.631 $\pm$ 0.014	0.631 $\pm$ 0.014	0.626 $\pm$ 0.014	0.627 $\pm$ 0.014	0.626 $\pm$ 0.014	-0.004 $\pm$ 0.020
		0.719 $\pm$ 0.000	0.719 $\pm$ 0.000	0.704 $\pm$ 0.001	0.713 $\pm$ 0.001	0.711 $\pm$ 0.001	-0.006 $\pm$ 0.001
	synthetic_c2	0.619 $\pm$ 0.002	0.626 $\pm$ 0.002	0.554 $\pm$ 0.000	0.634 $\pm$ 0.002	0.634 $\pm$ 0.002	+0.008 $\pm$ 0.003
		0.628 $\pm$ 0.003	0.630 $\pm$ 0.003	0.524 $\pm$ 0.003	0.633 $\pm$ 0.002	0.633 $\pm$ 0.002	+0.002 $\pm$ 0.005
	synthetic_c5	0.820 $\pm$ 0.012	0.817 $\pm$ 0.013	0.826 $\pm$ 0.004	0.818 $\pm$ 0.013	0.818 $\pm$ 0.013	+0.005 $\pm$ 0.018
		0.802 $\pm$ 0.006	0.804 $\pm$ 0.007	0.793 $\pm$ 0.007	0.804 $\pm$ 0.008	0.803 $\pm$ 0.008	-0.000 $\pm$ 0.010
xes3g5m	very_cold	0.783 $\pm$ 0.003	0.787 $\pm$ 0.003	0.745 $\pm$ 0.004	0.780 $\pm$ 0.003	0.776 $\pm$ 0.003	-0.007 $\pm$ 0.005
		0.708 $\pm$ 0.000	0.717 $\pm$ 0.000	0.606 $\pm$ 0.000	0.709 $\pm$ 0.000	0.703 $\pm$ 0.000	-0.009 $\pm$ 0.000

13 GKT epoch-extended ablation

Table S21 (Table 18) compares the released GKT configuration (10 epochs, batch 16) with a GKT-only epoch extension on split seed 42 (`configs/xes3g5m_gkt_epochs30.yaml`; 30 epochs, batch 32 under our 16–24GB GPU budget; *simpleKT* reference unchanged) on XES3G5M under train-only graphs. Fold-level Δ AUC intervals use paired-*t* over three learner-disjoint folds, matching Table S16.

14 Multi-seed GKT replication

Table S22 (Table 19) replicates GKT epoch-extended training at split base seeds 17, 42, and 1234. Each run rebuilds fold-specific train-only graph exports before training (see `configs/xes3g5m_split17.yaml` and siblings). Δ AUC pairs GKT against *simpleKT* from the matched Phase 3 trio runs at the same seed (GKT: 30 epochs, batch 32; *simpleKT*: 10 epochs, batch 64). Pooled nine-fold summary: Table 20.

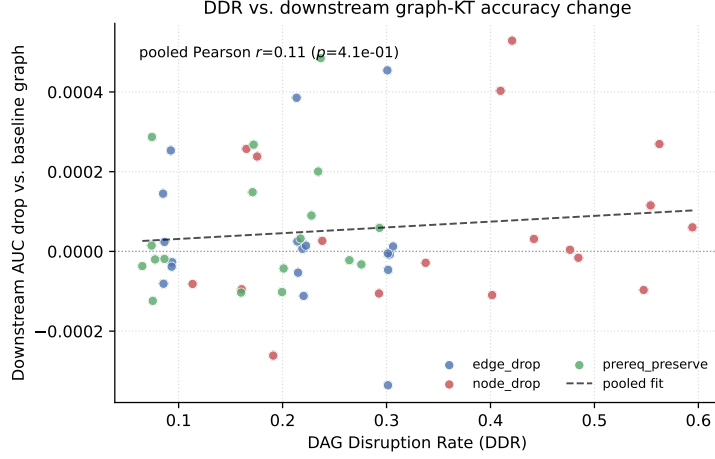


Fig. 2: Supplementary Figure S3. Downstream DGEKT sensitivity to prerequisite-graph disruption on ASSISTments 2012 and XES3G5M. Each point is one fold/operator/strength run; the vertical axis is AUC drop relative to the unperturbed train-only graph.

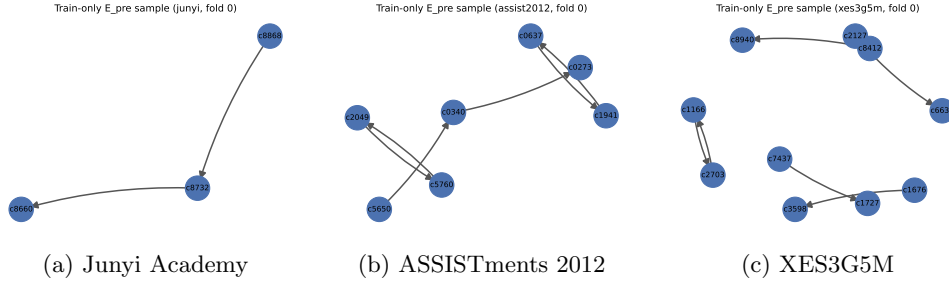


Fig. 3: Supplementary Figure S1. High-weight samples from train-only E_{pre} after DAG audit (fold 0 exports). The main manuscript shows the Junyi panel only.

15 Controlled Leak Injection

Table S17 (Table 21) probes the sensitivity of the audit metrics by deliberately injecting 5% and 20% of test-fold interactions into the train-only prerequisite builder on XES3G5M (fold 0). Edge counts $|E_{\text{pre}}|$ are *candidate* edges before DAG pruning (the main-text DAG audit reports 701 retained edges after pruning on the clean fold). Candidate-edge count and TBMR_f increase monotonically with the injection rate, while $\text{ECR}_f^{\text{flag}}$ remains 0 (learner-id sets stay disjoint) and $\text{ECR}_f^{\text{overlap}}$ is already saturated at 1.0 on this transition-dense corpus. Flag-level checks alone therefore miss this contamination channel, supporting the tiered audit design in the main manuscript.

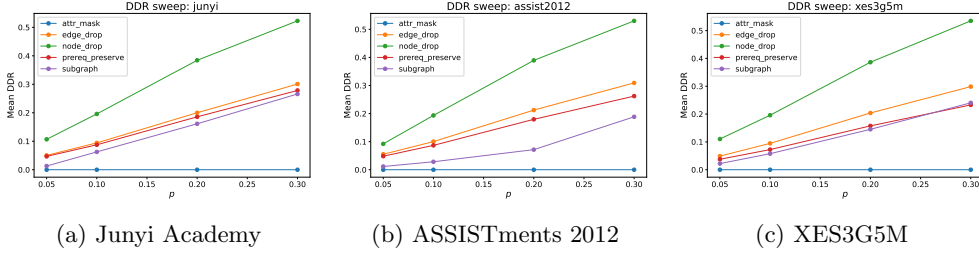


Fig. 4: Supplementary Figure S2. DDR versus perturbation strength p for five augmentation families across three benchmarks. The main manuscript shows the Junyi panel only.

Table 16: Training parity and hyperparameters (Table S15)

Model	Codebase	Graph input	Epochs	Batch	ES patience	Runtime	Hardware
BKT	scipy L-BFGS	—	30	64	N/A	<5 min	CPU
DKT	pyKT stock	—	30	64	5	30–180 min	A100/V100
<i>simpleKT</i>	pyKT stock	—	30	64	5	30–180 min	A100/V100
AKT	pyKT stock	—	30	64	5	30–180 min	A100/V100
GKT	pyKT stock	$E_{\text{pre}} + E_{\text{sim}}$	10	16	5	30–180 min	A100/V100
GIKT	native	$E_{\text{pre}} + E_{\text{sim}}$	10	16	5	30–180 min	A100/V100
SKT	native	$E_{\text{pre}} + E_{\text{sim}}$	10	16	5	30–180 min	A100/V100
DyGKT	native	$E_{\text{pre}} + E_{\text{sim}}$	10	16	5	30–180 min	A100/V100
DGEKT	native	$E_{\text{pre}} + E_{\text{sim}}$	10	16	5	30–180 min	A100/V100

Epoch-matched GKT ablation (Tables S21–S22): `configs/xes3g5m.gkt.epochs30.yaml` (30 epochs, batch 32; *simpleKT* reference runs in S22 retain batch 64 from Phase-3 trio reruns).

Table 17: ΔAUC intervals for primary pairs, XES3G5M, train-only graphs (Table S16)

Model Pair	ΔAUC	95% CI (paired t , 3 folds)	n_{learners}	n_{rows}
GKT vs <i>simpleKT</i>	−0.041	[−0.044, −0.038]	—	—
GIKT vs <i>simpleKT</i>	+0.003	[+0.003, +0.003]	—	—

Table S18 (Table 22) reports fold-0 test AUC after retraining *simpleKT*, GKT, and GIKT on the clean and injected graphs. Headline AUC does not track the structural audit monotonically: GKT and GIKT AUC *rise* at 20% injection on this dense log, whereas *simpleKT* remains near 0.850–0.858. We therefore treat S18 as evidence that metric-tier reporting must precede AUC-only leakage conclusions, not as proof that contamination always lowers accuracy.

Table 18: GKT epoch-extended ablation on XES3G5M, split seed 42 (Table S21)

GKT config	Epochs	Batch	Mean AUC	Δ vs <i>simpleKT</i>	95% CI (3 folds)
Primary release	10	16	0.834	-0.041	$[-0.044, -0.038]$
Epoch-matched ablation	30	32	0.837	-0.038	$[-0.048, -0.027]$
<i>simpleKT</i> reference	30	64	0.875	—	—
Epoch gain (30ep - 10ep GKT)			+0.003		$[-0.004, +0.011]$

Table 19: GKT epoch-extended vs. *simpleKT* by split seed (Table S22)

Tag	Seed	Folds	Δ AUC (GKT - <i>simpleKT</i>)	Fold deltas
gkt_epochs30_s17	17	3	-0.034 ± 0.000	-0.034367; -0.034458; -0.033971
gkt_epochs30_s42	42	3	-0.038 ± 0.004	-0.034211; -0.036040; -0.042334
gkt_epochs30_s1234	1234	3	-0.041 ± 0.003	-0.042178; -0.043039; -0.037002

Table 20: Pooled nine-fold Δ AUC summary (Table S22, continued)

Statistic	Value
Pooled Δ AUC (GKT - <i>simpleKT</i>)	-0.038 [-0.040, -0.035] (9 folds; paired- <i>t</i>)
Split base seeds	17, 42, 1234 (fold-aligned graph rebuild each)

Table 21: Controlled Leak Injection, XES3G5M fold 0. $|E_{\text{pre}}|$ counts candidate edges before DAG pruning. (Table S17)

Injection rate	$ E_{\text{pre}} $	ECR ^{overlap}	ECR ^{flag}	TBMR
0%	1162	1.000	0.000	0.754
5%	1378	1.000	0.000	0.776
20%	1417	1.000	0.000	0.779

16 Exploratory ANOVA summaries

The main manuscript reports fold-level means $\pm$ std throughout Tables 7, 8, 5, 10, and 12, with Table S16 Δ AUC intervals (Table S16) as the primary inferential evidence for the XES3G5M ranking observation. For reviewer-requested completeness, Tables S19–S20 add exploratory ANOVA models on fold-level observations. Because each dataset–model cell contains only three folds, we interpret ANOVA *p*-values as descriptive rather than as proof of backbone dominance.

Table 22: Downstream test AUC under controlled leak injection, XES3G5M fold 0 (Table S18)

Model	Clean	5% leak	20% leak
<i>simpleKT</i>	0.850	0.852	0.858
GKT	0.810	0.825	0.860
GIKT	0.852	0.860	0.880

Table 23: Exploratory two-way ANOVA on fold-level AUC ($n=3$ folds per dataset-model cell; public benchmarks only, train-only graphs; Table S19). Partial $\eta^2 = SS_{\text{effect}}/(SS_{\text{effect}} + SS_{\text{error}})$. With only three folds per cell, p -values are underpowered; fold-level ΔAUC magnitudes and Table S16 intervals are the primary evidence.

Source	SS	df	MS	p	η_p^2
Dataset	0.3365	2	0.1682	<0.001	1.000
Backbone	0.0701	5	0.0140	<0.001	0.998
Dataset $\times$ Backbone	0.0968	10	0.0097	<0.001	0.999
Residual	0.0001	34	0.0000	—	—

Table 24: Exploratory ANOVA on DGEKT downstream AUC under graph perturbation (ASSISTments and XES3G5M; three folds per operator/strength; Table S20). Dependent variable: test AUC after retraining on the perturbed graph. Partial η^2 as above. This analysis supports the null predictive-sensitivity finding in Section 4.7; it does not establish backbone dominance.

Source	SS	df	MS	p	η_p^2
Dataset	0.1893	1	0.1893	<0.001	1.000
Operator	0.0000	2	0.0000	0.973	0.001
Operator $\times$ strength p	0.0000	6	0.0000	0.996	0.014
Dataset $\times$ Operator	0.0000	2	0.0000	0.890	0.006
Residual	0.0000	42	0.0000	—	—